Exercise5

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Contenido

Exercise 5

Use the `pandas.crosstab` to find solution to the problem below.

	Credit Card	Cash	Mobile Pay	Total
Under $50	85	120	45	250
50—100	110	40	50	200
Over $100	40	5	5	50
Total	235	165	100	500

a. What is the probability that a randomly selected transaction is over $100?

b. What is the probability that a transction uses Cash?

c. What is the probability that a transaction is between 50—100 AND uses Mobile Pay?

d. If a transaction is under $50, what is the probability it was paid with Credit Card?

e. If a transaction uses Cash, what is the probability it is under $50?

(Answers: a. 0.10 b. 0.33 c. 0.10 d. 0.34 e. 0.727)

In [4]:

```
#Solution here
import pandas as pd
data = {
    'Amount_status' : ['Under $50']*250 + ['$50 - $100']*200 + ['Over $100']*50,
    'Payment_status' :(['Credit Card']*85 + ['Cash']*120 + ['Mobile Pay'] * 45 +
                       ['Credit Card']*110 + ['Cash']*40 + ['Mobile Pay'] * 50 +
                       ['Credit Card']*40 + ['Cash']*5 + ['Mobile Pay'] * 5)
}
df = pd.DataFrame(data)
# Create contingency table
table = pd.crosstab(df['Amount_status'], df['Payment_status'], margins=True)
print(table)

total = table.loc['All','All']
print(" ")
```

Out [4]:

```
Payment_status  Cash  Credit Card  Mobile Pay  All
Amount_status
$50 - $100        40          110          50  200
Over $100          5           40           5   50
Under $50        120           85          45  250
All              165          235         100  500
```

(a) Probability that a transaction is over $100

$$P(\text{Over } 100) = \frac{50}{500} = 0.10$$

In [39]:

```
# probability that a randomly selected transaction is over $100
print(f"P(Over $100) = {table.loc['Over $100','All']/total:.2f}")
```

Out [39]:

```
P(Over $100) = 0.10
```

(b) Probability that a transaction uses Cash

$$P(\text{Cash}) = \frac{165}{500} = 0.33$$

In [11]:

```
# probability that a transction uses Cash?
print("P(Cash) =",table.loc['All','Cash']/total)
```

Out [11]:

```
P(Cash) = 0.33
```

(c) Probability that a transaction is between $50–\$100 AND uses Mobile Pay

$$P(50\text{–}100 \cap \text{Mobile Pay}) = \frac{50}{500} = 0.10$$

In [40]:

```
#probability that a transaction is between  $50-$100 AND uses Mobile Pay
print(f"P($50-$100 and Mobile Pay) =  {table.loc['$50 - $100','Mobile Pay']/total:.2f}")
```

Out [40]:

```
P($50-$100 and Mobile Pay) =  0.10
```

(d) If a transaction is under $50, probability it was paid with Credit Card

$$P(\text{Credit Card} \mid \text{Under } 50) = \frac{85}{250} = 0.34$$

In [30]:

```
#If a transaction is under $50, probability that a transaction is paid with Credit Card
print("P(Credit Card given Under $50) =", table.loc['Under $50','Credit Card']/table.loc['Under $50','All'])
```

Out [30]:

```
P(Credit Card given Under $50) = 0.34
```

(e) If a transaction uses Cash, probability it is under $50

$$P(\text{Under } 50 \mid \text{Cash}) = \frac{120}{165} \approx 0.727$$

In [37]:

```
# If a transaction uses Cash, the probability that is under $50?
print(f"P(Under $50 given Cash) = { table.loc['Under $50','Cash']/table.loc['All','Cash']:.3f}")
```

Out [37]:

```
P(Under $50 given Cash) = 0.727
```